

Ji-Seong Han

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Data Scientist with experience in machine learning, large-scale data processing, and applied AI, with hands-on experience in LLMs and computer vision. Proven ability to build predictive models and data pipelines that translate complex data into actionable insights.

Education

MS Data Science (Computing & Decision Analytics)

Expected Graduation: December 2026

- Cumulative GPA 3.70, Arizona State University, Tempe, AZ

BS Industrial and Information System Engineering

March 2018 – August 2024

- Cumulative GPA 3.52, Soongsil University, Seoul, South Korea

Technical Skills

Programming: Python, R, SQL

ML/DL: PyTorch, TensorFlow, Transformers (LLMs), Computer Vision, XGBoost, LightGBM

Data Analysis & Processing: Pandas, NumPy, Spark, Kafka

MLOps / Infra: Docker, Kubernetes (Minikube), Neo4j, Git

Databases: MySQL, PostgreSQL

Project Experience

Reproduction of RVT: Recurrent Vision Transformers for Object Detection with Event Cameras

May 2026

Computer Vision, Event-based Vision, PyTorch

- Reproduced and validated a CVPR 2023 event-based object detection model using the Gen1 dataset on GPU environments
- Achieved 47.15% mAP, closely matching the reported performance (47.2%), validating reproducibility
- Implemented a hybrid architecture combining Transformer-based spatial modeling and LSTM-based temporal aggregation
- Conducted GPU-based experiments to analyze model efficiency, inference behavior, and reproducibility
- Leveraged ASU Sol supercomputing environment to run large-scale experiments, validating model performance and reproducibility

SafeCase – AI-Powered Nonprofit Case Management Platform | Best Security Build

March 2026

Full-Stack Development, AI Integration, System Design

- Built an AI-powered case management platform adopted by multiple nonprofit organizations, replacing spreadsheet-based workflows
- Designed and deployed full-stack system using Next.js, Supabase (PostgreSQL), Vercel, enabling real-time client and service tracking
- Implemented Google SSO and role-based access control (RBAC) to ensure secure multi-user access across Admin and Staff roles
- Integrated LLM-based features (Claude API) including document parsing (photo-to-intake), automated follow-up detection, and natural language querying over client records
- Reduced operational cost from \$50–150/user/month to under \$30/month through a lightweight, scalable architecture

Text-to-SVG Generation using Large Language Models

June 2025

Large Language Models, PEFT/QLoRA, LoRA, Prompt Engineering, CLIP Evaluation

- Fine-tuned open-source LLMs including Mistral-7B, LLaMA, T5, Gemma, and Qwen-Coder for text-to-SVG code generation using the starvector/text2svg-stack dataset
- Built an end-to-end training pipeline covering dataset filtering, SVG normalization, tokenizer-based formatting, and chat-style prompts
- Applied LoRA, QLoRA, 4-bit quantization, and gradient checkpointing to fine-tune multi-billion-parameter models under limited GPU resources
- Achieved 73% valid SVG generation, 71% successful rasterization, and an average CLIP score of 0.274 on 100 unseen prompts
- Compared multiple model architectures and prompting strategies to improve generation quality and consistency

Predictive Quality Management System | Grand Prize

July 2024

Machine Learning, Predictive Modeling, Pseudo-Labeling

- Developed a predictive maintenance model for injection-molding processes using real manufacturing data
- Designed a pseudo-labeling strategy to address missing defect labels in industrial datasets
- Increased production yield (~73% → ~79%) and reduced defect rate (~16% → ~1.5%)
- Built a hybrid pipeline combining traditional quality control methods with machine learning models
- Awarded Grand Prize for delivering a practical, deployable AI solution with measurable business impact

Professional Experience

Tableau Bootcamp Trainee

October 2024 – November 2024

- Completed hands-on training in Tableau fundamentals, core visualizations, and dashboard design
- Built interactive dashboards using real datasets for business insight generation

LG Aimers Data Intelligence Bootcamp

January 2024 – February 2024

- Built B2B sales opportunity prediction models using LightGBM, XGBoost, CatBoost, RandomForest as a 4-member team
- Engineered features from 100k+ MQL records by incorporating domain-specific attributes (lead source, activity logs)
- Applied undersampling + subset-based ensemble training to improve recall on highly imbalanced data
- Ranked 63rd out of 844 teams, demonstrating strong modeling performance under competitive evaluation

LG HelloVision DX Data School

June 2023 – December 2023

- Trained in Python preprocessing, SQL/NoSQL modeling, Tableau visualization, and ML fundamentals
- Developed a personalized VOD recommendation system using real user-log data, incorporating behavioral feature engineering and custom collaborative-filtering models
- Engineered user-behavior features and refined pipelines through collaborative code reviews
- Designed and evaluated multiple ML approaches and created dashboards enabling business teams to interpret user patterns